RYAN GAO

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PROFESSIONAL SUMMARY

Data analyst with a quantitative foundation in Astrophysics from UC Berkeley and credentials in Data Science and Full-Stack Development. Experienced in dataset annotation, model evaluation, Python ETL pipelines, and digital signal processing. Focuses on empirical verification, reproducible workflows, and clean technical documentation.

TECHNICAL SKILLS

- **Languages & Core Tools:** Python (Pandas, NumPy, GeoPandas), SQL+PostgreSQL, JavaScript, HTML/CSS, Git/GitHub
- **Data Operations & ML:** Data Annotation, RLHF Evaluation, Taxonomy Curation, Prompt Testing, Model Auditing
- **Engineering & Tooling:** ETL Pipelines, Data Sanitization, REST APIs, Docker, MongoDB, Folium/Leaflet
- **Quantitative Methods:** Statistical Modeling, Numerical Analysis, Signal Processing, Fourier Transform, Linear Algebra

PROFESSIONAL EXPERIENCE

Data Specialist & Evaluation Contractor | Workada (Remote) | 08/2026 – Present

- Evaluate and compare paired model outputs across diverse technical and domain-specific prompts.
- Author structured qualitative justifications detailing comparative model strengths, flaws, and trade-offs.
- Audit model responses against source materials to identify factual errors, logical flaws, and edge-case failures.
- Apply multi-tier quality rubrics to maintain consistent evaluation benchmarks across subject areas.

Quantitative Instructor (Mathematics & Physics) | Elite Education (Hong Kong) | 09/2024 – 10/2025

- Delivered analytical instruction across multivariable calculus, linear algebra, trigonometry, and classical mechanics.
- Deconstructed complex mathematical proofs into structured, reproducible problem-solving frameworks.
- Maintained detailed diagnostic progress records, tracking quantitative benchmarks and learning outcomes.

Volunteer Data Analyst | Hack for LA (Los Angeles, CA) | 10/2019 – 02/2020

- Built Python and Pandas ETL pipelines to ingest, clean, and process high-volume municipal 311 service call datasets.
- Contributed to backend service endpoints using Sanic and assisted with containerization using Docker.

Research Intern (Signal Processing) | Breakthrough Listen / UC Berkeley (Berkeley, CA) | 05/2016 – 08/2016

- Analyzed radio frequency interference (RFI) patterns by evaluating ground-level radio signals against baseline observation survey data.
- Wrote Python scripts to isolate candidate signals, perform Fast Fourier Transforms (FFT), and produce spectral plots.
- Documented work in Jupyter notebooks as reference for future interns and students.

KEY PROJECTS

Redistricting Tracker | Python, GeoPandas, Folium, JavaScript

- Engineered an interactive geospatial tool tracking congressional districts altered by mid-decade redistricting.
- Ingested boundary geometries and precinct-level election returns to benchmark partisan margin shifts and net swing against baseline results.
- Built responsive choropleths and hover inspection tooltips displaying district statistics and margin deltas.

LA 311 Civic Analytics Pipeline | Python, Pandas, PostgreSQL, Docker

- Contributed to an open-source civic analytics pipeline transforming Los Angeles' 311 logs into structured metrics.
- Developed automated data cleaning routines and assisted with relational database normalization.

EDUCATION

University of California, Berkeley | Bachelor of Arts in Astrophysics | 2011 – 2016

- Key Coursework: Multivariable Calculus, Linear Algebra, Radio Astronomy Laboratory, Quantum Mechanics

MITx Online | MicroMasters Credential in Data Science | 2018 – 2019

- Key Coursework: Probability & Statistics, Data Analysis for Social Scientists, Machine Learning with Python

UCLA Extension | Certificate in Full-Stack Web Development | 2018 – 2019

- Key Coursework: Database Architecture, Relational SQL, Node.js, React.js

ADDITIONAL

- U.S. Citizen
- Languages: English (Native), Mandarin (Colloquial)